AMATH 353 HW 6

Jordan Tucker

May 23, 2026

Question 1: Method of Characteristics (part 1

(a) Find the characteristic curves of the PDE

$$u_t + xu_x = 0.$$

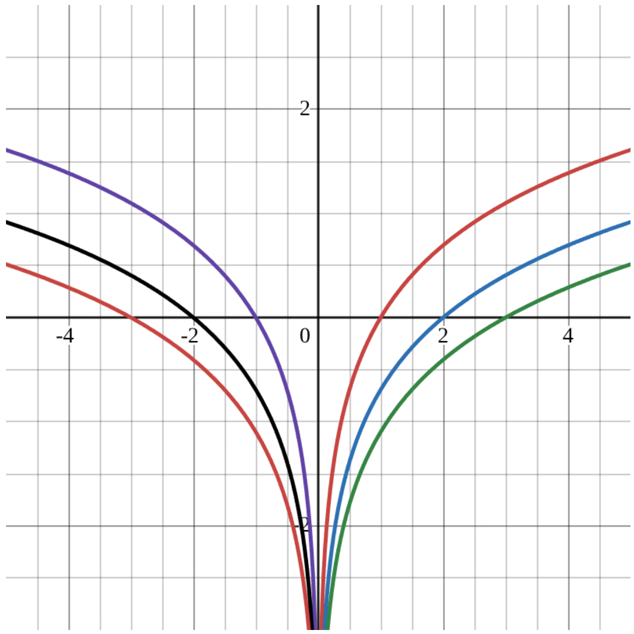
Sketch/plot the characteristic curves in the xt-plane. What do you predict the dynamics tolook like?

$$\begin{gathered}
u_t + xu_x = 0 \\
\frac{d}{dt}u(x(t), t) = u_t + \frac{dx}{dt}u_x \\
u_t + \frac{dx}{dt}u_x = u_t + xu_x \\
\frac{dx}{dt} = x
\end{gathered}$$

Solve the ODE:

$$\begin{gathered}
\frac{1}{x}dx = dt \\
\int \frac{1}{x}dx = \int dt \\
\ln|x| = t + \ln|x_0| \\
x = e^{t+\ln|x_0|} \\
x = x_0 e^t
\end{gathered}$$

The wave will go faster the further it gets from the origin. If $x_0 > 0$ it moves to the right. If $x_0 < 0$ it moves to the left.



(b) Use the method of characteristics to solve

$$\begin{cases} u_t + tu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = \cos(x), & x \in \mathbb{R}, t \geq 0. \end{cases}$$

Give a qualitative description of the dynamics of the solution.

$$u_t + tu_x = 0$$
$$\frac{d}{dt}u((x),t) = u_t + \frac{dx}{dt}u_x$$
$$u_t + tu_x = u_t + \frac{dx}{dt}u_x$$
$$\frac{dx}{dt} = t$$

Solve the ODE

$$\frac{dx}{dt} = t$$
$$dx = tdt$$
$$\int dx = \int tdt$$
$$x = \frac{t^2}{2} + x_0$$
$$x_0 = x - \frac{t^2}{2}$$
$$u(x_0, 0) = \cos(x - \frac{t^2}{2})$$

$$u(x, t) = \cos(x - \frac{t^2}{2})$$

The wave speeds up as it moves away from x_0, and always moves to the right.

(c) Use the method of characteristics to solve

$$\begin{cases} u_t + \cos(t)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = e^{-x^2}, & x \in \mathbb{R}, t \geq 0. \end{cases}$$

Give a qualitative description of the dynamics of the solution.

$$u_t = \cos(t)u_x = 0$$
$$\frac{d}{dt}u(x(t), t) = u_t + \frac{dx}{dt}u_x$$
$$u_t + \frac{dx}{dt}u_x = u_t + \cos(t)u_x$$
$$\frac{dx}{dt} = \cos(t)$$

Solve the ODE

$$\frac{dx}{dt} = \cos(t)$$
$$dx = \cos(t)dt$$
$$\int dx = \int \cos(t)dt$$
$$x = \sin(t) + x_0$$
$$x_0 = x - \sin(t)$$

$$u(x_0, 0) = e^{-(x-\sin(t))^2}$$
$$u(x, t) = e^{-(x-\sin(t))^2}$$

The wave bounces between -1 and 1, and the velocity depends on time.

Question 2: Method of Characteristics (part 2)

(a) Determine the (implicit) solution to the nonlinear, homogeneous PDE problem

$$\begin{cases} u_t + u^2 u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = \tanh(x), & x \in \mathbb{R}. \end{cases}$$

$$\begin{gathered} x = x(t) \\ \frac{d}{dt}u(x(t),t) = u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x = u_t = u^2 u_x \\ \frac{dx}{dt} = u^2 \end{gathered}$$

Solve The ODE

$$\begin{gathered} \frac{dx}{dt} = u^2 \\ dx = u^2 dt \\ \int dx = \int u^2 dt x = u^2 t + x_0 \\ x_0 = x - u^2 t \\ u(x_0, 0) = \tanh(x_0) \\ u(x,t) = \tanh(x_0) \\ u(x,t) = \tanh(x - u^2 t) \end{gathered}$$

(b) Determine the (explicit) solution to the linear, in-homogeneous PDE problem

$$\begin{cases} u_t + 2t u_x = 2t\cos(x), & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = \operatorname{sech}^2(x), & x \in \mathbb{R}. \end{cases}$$

$$\begin{gathered} \frac{d}{dt}u(x(t),t) = u_t + \frac{dx}{dt}u_x \\ u_t + \frac{dx}{dt}u_x = u_t 2t + u_x \\ \frac{dx}{dt} = 2t \end{gathered}$$

Solve the ODE

$$\frac{dx}{dt} = 2t$$
$$dx = 2tdt$$
$$\int dx = \int 2tdt$$
$$x = t^2 + x_0$$

$$\frac{du}{dt} = 2t\cos(x(t))$$
$$\int du = \int 2t\cos(t^2 + x_0)dt$$
$$u(t) = \sin(t^2 + x_0) + C$$

Initial Conditions

$$t = 0,\ u(x,0) = \operatorname{sech}^2(x)$$
$$t = 0,\ x(0) = x_0$$
$$t = 0,\ u(0) = \sin(x_0) + C$$
$$\sin(x_0) + C = \operatorname{sech}^2(x_0)$$
$$C = \operatorname{sech}^2(x_0) - \sin(x_0)$$

$$u(x_0, t) = \sin(t^2 + x_0) + \operatorname{sech}(x_0) - \sin(x_0)$$

$$x = t^2 + x_0$$
$$x_0 = x - t^2$$
$$u(x,t) = \sin(t^2 + (x - t^2)) + \operatorname{sech}^2(x - t^2) - \sin(x - t^2)$$

(c) Determine the (explicit) solution to the non-linear, in-homogeneous PDE problem

$$\begin{cases} u_t + uu_x = \cos(t), & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = x, & x \in \mathbb{R}. \end{cases}$$

Question 3: Breaking Time

The breaking time t_b is the first (non-negative) time at which characteristics intersect, and we computed

$$t_b = \min_{x_0 \in \mathbb{R}} \left\{ \frac{-1}{c'(u_0(x_0))u_0'(x_0)} \geq 0 \right\} \quad \text{for} \quad \begin{cases} u_t + c(u)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}. \end{cases}$$

For each of the following problems, determine whether the solutions breaks and if so, compute the breaking point (x_b, t_b).

(a) The non-viscous traffic model $u_t + v_1(1 - \frac{u}{u_1})u_x = 0$ with initial profile $u_0(x) = \frac{u_1}{2}e^{-(x/\beta)^2}$. How does β impact the initial profile and the breaking time?

(b) The wacky PDE $\quad u_t + \sinh\left(\sqrt{1 + \exp(u)}\right)u_x = 0$ with initial profile $u_0(x) = \tanh(x)$.

(c) Find a PDE $u_t + c(x,t)u_x = 0$ and initial profile $u_0(x)$ so that the solution becomes arbitrarily steep as $t \to +\infty$ but never breaks.

Question 4: Understanding Qualitative Behavior

In this problem, you will (rigorously) establish several results concerning the qualitative behavior of solutions to large classes of conservation laws. Consider the PDE problem given by

$$\begin{cases} u_t + c(x,t,u)u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}. \end{cases}$$

You may assume that $c(x,t,u)$ and $u_0(x)$ are smooth functions, i.e., you can safely differentiate them.

(a) Prove that if $c = c(t)$, then the profile never deforms, i.e., $u(x,t)$ is a shift of the initial condition.

(b) Prove that if $c = c(x) > 0$, then the solution never breaks, regardless of $u_0(x)$.

(c) Prove that if $c = c(u) \neq \text{const.}$, then there exists an initial profile $u_0(x)$ corresponding to a solution which breaks in finite time.

(d) Prove that if $c = c(u)$ and $u_0(x)$ are both increasing functions, then the solution will never break.